Tool-Aware Ergodic Stroke Planning and Voxel Simulation for Robotic Sgraffito

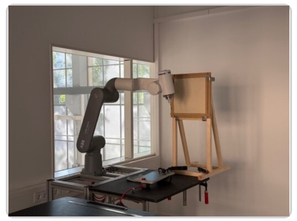
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Scan for clips and more

Image-to-carve planning for robotic sgraffito: task decomposition, ergodic stroke placement under tool-footprint constraints, and voxel-simulated verification.

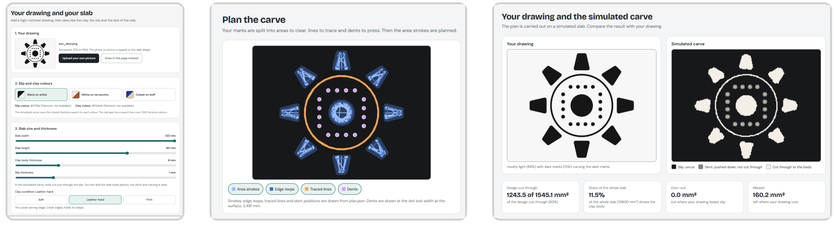
Prepare (input image) → **Design** (artist edits) → **Rules** (carving rules) → **Decompose** (regions & layers) → **Plan** (tool paths) → **Simulate** (carve preview) → **Render** (final output)



1 Background: sgraffito

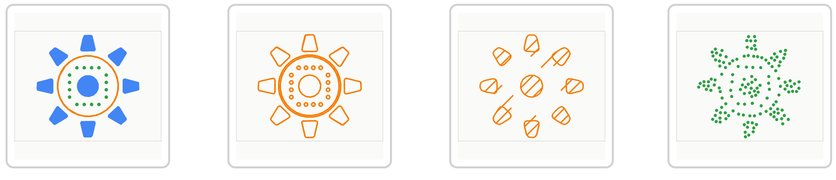
Sgraffito is a decorative technique used in Renaissance architecture and ceramics, where carving through a top layer reveals a contrasting colour underneath. We're exploring how robots can create these designs, with simulation allowing artists to sketch ideas digitally and preview the results before carving them into the material.

2 Artist Interface



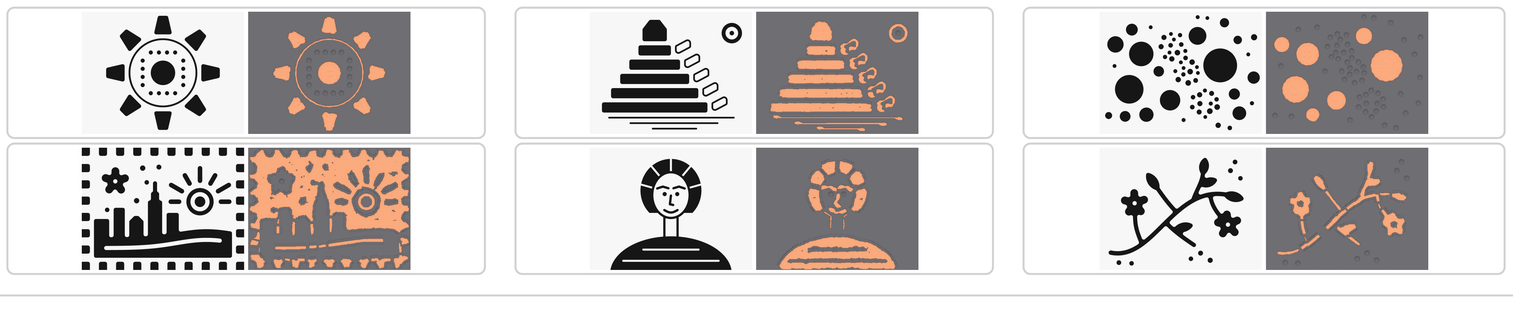
Web interface: the artist uploads a drawing and picks a slab, plans the carve, then views the simulated result before the robot runs.

3 Image Stylization

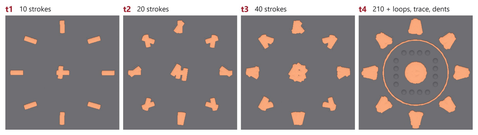


Four stylizations: regions, outlines, hatching, stipple. All become one shape format.

4 Input vs. voxel-simulated carve (0.25 mm voxels, Blender render).



5 Ergodic fill over time



Coverage $C(N)$ after $N = 10, 20, 40, 210$ strokes, taking strokes in turn from each region: 28% → 44% → 60% → 76%. Gains fall from about 1.6% to under 0.1% per stroke, so $C(40) \approx 0.8\,C(210)$.

$$\phi = \mathbf{1}_R / |R|$$

$$\phi_k = \int_R \phi(x)\, f_k(x)\, dx$$

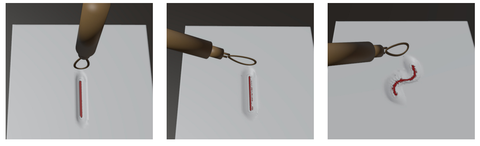
$$c_k = \frac{1}{N} \sum_{j=1}^{N} \bar{f}_k(\sigma_j)$$

$$E = \sum_k \Lambda_k \left(c_k - \phi_k\right)^2$$

$$\Lambda_k = \left(1 + \|k\|^2\right)^{-3/2}$$

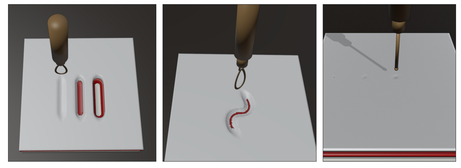
$$\sigma_{N+1} = \arg\min_{\sigma \subset R} E(\sigma_{1..N}, \sigma)$$

6 Tool Tilt



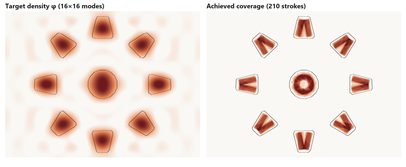
Loop tool at $\theta = 0°, 30°, 60°$ along a curve. Each tool is modelled by its removal mesh M: through-cut width $w(d) = \text{width}_x(M \cap \{z = d - s\})$ for cut depth d and slip thickness s. At equal d the footprint stays similar; θ mainly changes the approach angle.

7 Action Primitives



Scrape $\varphi_\ell \leftarrow \max(\varphi_\ell, -\varphi_M \circ g^{-1})$
Trace same cut, g along a polyline
Press $\dot{\varepsilon}_p = r\langle p/p_y - 1\rangle^n,\ h = h_0(1 - \varepsilon_p)$

8 Coverage



Left: target density ϕ seen through 16×16 Fourier modes. Right: achieved stroke coverage. Strokes stay inside each shape and concentrate where ϕ is high.

Carve Accuracy

Design	Cut	Over	Missed
Circles	90.9%	0	43.5
Branch	62.9%	0	346.5
Portrait	74.5%	48	432.2
Pyramid	92.4%	168	222.2
Mixed	75.5%	0	1023.8
Sun	80.5%	0	160.2

Simulated. Cut = % of design cut through (includes over-cut). Over, Missed in mm^2.

References. [1] G. Mathew & I. Mezić, *Metrics for ergodicity and design of ergodic dynamics for multi-agent systems*, Physica D 240(4–5):432–442, 2011. [2] E. Heiden, M. Macklin, Y. Narang, D. Fox, A. Garg & F. Ramos, *DiSECt: A differentiable simulation engine for autonomous robotic cutting*, RSS, 2021. [3] K. Museth, *VDB: High-resolution sparse volumes with dynamic topology*, ACM Trans. Graphics 32(3), 2013. [4] Blender Foundation, *Blender*, blender.org.

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